Figure M — Mechanism: richness × addressability

Matched dense-LIF forward, n = 20 per cell, 2026-08-25 re-run at MATCHED_INPUT_SCALE = 2.0. Pass / fail / at-chance — not a graded surface.

SuperSpike BPTT reference saturates: 1.0000 ff / 1.0000 rec, every suite, both graphs. Every other rule tested clears the gate, so this panel shows WHICH SINGLE RULE FAILS a task the rest saturate — it does not rank the rest.

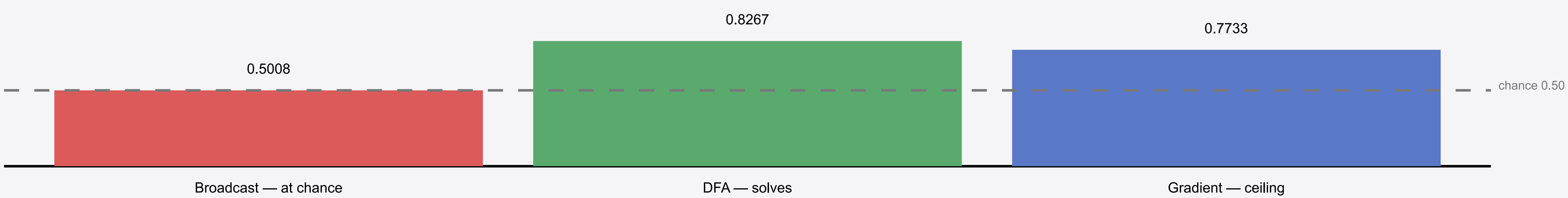
	Low addressability (broadcast)	High addressability (directed / local feedback)
Low richness (±1)	FAIL — **±1 × surrogate eligibility** MatchedLocal — the lead FAIL, and at chance on both graphs 0.5000 ff / 0.5100 rec contrast (not gated) — **±1 broadcast REINFORCE** MatchedRlFlat — same reward, same topology, well above chance 0.7775 ff / 0.7962 rec One cell, two rules, 0.28 apart. Collapsing them into one “broadcast ±1” box is the overreach the lead claim’s wording exists to avoid.	PASS — **REINFORCE × frozen B_i** MatchedRlReinforceFb — directed feedback, same ±1 reward 0.9950 ff / 0.9812 rec
High richness (graded)	contrast (not gated) — **broadcast graded error** MatchedBroadcastGradedError — on the DFA schedule; not a locality proof 0.9975 ff / 0.9975 rec	PASS — **graded error × DFA** MatchedDfaGradedError 0.9925 ff / 0.9875 rec

Richness alone does not decide it: a graded BROADCAST rule reaches 0.9975.

Addressability alone does not either: ±1 through a frozen B_i passes.

What fails is the one rule that has neither.

Panel B — XOR locality flip (1-layer xor_thresh). Addressability evidence: broadcast fails a task DFA solves.



Not claimed for 2-layer mid-init depth locality, where broadcast also solves. Matched PASS does not imply live muted-θ / k-WTA G2 PASS.